Supplementary Material for: The Hidden Privacy Cost of Valuing Hospitals in Federated Learning: Accounting, a Decision Rule, and Evidence from Real Federations

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This supplement contains additional methods, the reproduction of the base paper, the full analyses of the federation that is valued (federated training, differential privacy, transport and calibration; summarised in Section VI-A of the main paper), detailed tables, and all claims with verdicts computed from the result files. All analyses use real hospitals as released (the 19 PCa-19 hospitals and the four UCI Heart Disease hospitals); no hospital is simulated and no data set is partitioned artificially. Numbers without the prefix "S" refer to the main paper.

S1. Additional Methods

A. Subsampled Accounting

As an alternative we also train with Poisson subsampling: in every local step each patient is included independently with probability $q < 1$, the clipped gradients of the included patients are summed, noise $\mathcal{N}(0, \sigma^2 C^2 I)$ is added and the sum is divided by the *expected* batch size $q\, n_k$; $1/q$ steps make one epoch, so $N = T\,E/q$. One step is then a sampled Gaussian mechanism, whose RDP at integer order α is [1]–[3]

$$\epsilon_{RDP}(\alpha) = \frac{1}{\alpha - 1} \ln \sum_{j=0}^{\alpha} \binom{\alpha}{j} (1-q)^{\alpha-j} q^j \exp\Big(\frac{j^2 - j}{2\sigma^2}\Big), \tag{S1}$$

composed over the N steps and converted as above (orders 2–1024). For $q = 1$ it reduces to $\alpha/(2\sigma^2)$. Our implementation matches a numerical evaluation of the Rényi divergence to 10^{-13} and reproduces the value $\varepsilon \approx 1.26$ reported by Abadi *et al.* [4] for $q = 0.01$, $\sigma = 4$, 10,000 steps and $\delta = 10^{-5}$. Subsampling lowers the noise multiplier required for a given ε (privacy amplification [5]) at the price of more, smaller-batch steps; whether this improves utility is an empirical question (Section S2-H).

B. Membership Inference Attack Formulation

We audit empirical leakage with a confidence-threshold membership-inference attack. The score of a record is the model's confidence in its true label, $s_i = y_i p_i + (1-y_i)(1-p_i)$. Members (training patients) are subsampled to the number of non-members (test patients) and the attack is summarised by its ROC AUC. As a positive control, the same attack is applied to a deliberately over-fitted model (200 random noise features, no regularisation).

TABLE S1
Reproduction of the base paper's protocol on PCa-19 (50 Monte-Carlo repeats; AV = mean over hospitals, AVW = patient-weighted). Our re-implementation gives the same numbers as the authors' code (Section S2-A).

	Noh variables		Ettala variables	
Strategy	AV	AVW	AV	AVW
Published calculator (BL)	0.827	0.847	0.792	0.830
Local (LOC)	0.818	0.847	0.799	0.847
Centralised (CEN)	0.827	0.849	0.813	0.852
One-shot FedAvg (FL)	0.826	0.848	0.803	0.843
Free-rider (FR)	0.827	0.847	0.803	0.840

S2. Additional PCa-19 Analyses

This section gives the full analyses of the federation that is valued, summarised in Section VI-A of the main paper.

A. Reproduction of the Authors' Protocol

We re-implemented the authors' public protocol line by line: scikit-learn logistic regression on the raw Ettala and Noh variables, a stratified 80/20 split per hospital, one-shot FedAvg (a sample-size-weighted average of fully trained local models), and free-riding evaluated on 100% of the excluded hospital. Monte-Carlo random states are drawn exactly as in their scripts. Running the authors' own code and our re-implementation with the same seed gives identical matrices for both calculators (largest difference 1.2×10^{-14}; a validation check reruns this comparison). Table S1 confirms the base paper's qualitative findings: the published Noh calculator is almost as good as centralised training (AV 0.827 vs 0.827), the Ettala architecture is weaker, and FL, FR and CEN are within a few thousandths of each other.

B. Training Strategies under Our Protocol

Under our protocol (standardised log-PSA, log-volume, age and PI-RADS indicators; iterative FedAvg/FedProx with $T = 20$, $E = 5$; 20 seeds), the sustainability result of the base paper holds on its own data (Fig. S1). Joining the federation gives no significant average gain over local training (FL$-$LOC $= -0.005$ $[-0.015, +0.006]$). A free-rider obtains almost the same model (FR$-$FL $= -0.002$ $[-0.002, -0.001]$), and federated training matches pooled training (CEN$-$FL $= +0.002$ $[-0.001, +0.004]$). Per hospital, 4 hospitals gain

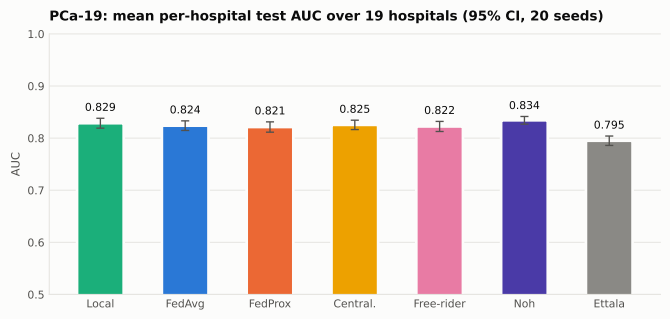


Fig. S1. PCa-19: mean per-hospital test AUC of each strategy over the 19 real hospitals (20 seeds, 95% CI).

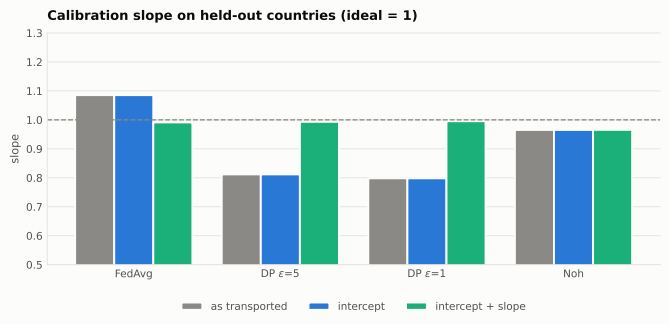


Fig. S2. Calibration slope on held-out countries before and after local recalibration on 20% of the country's patients.

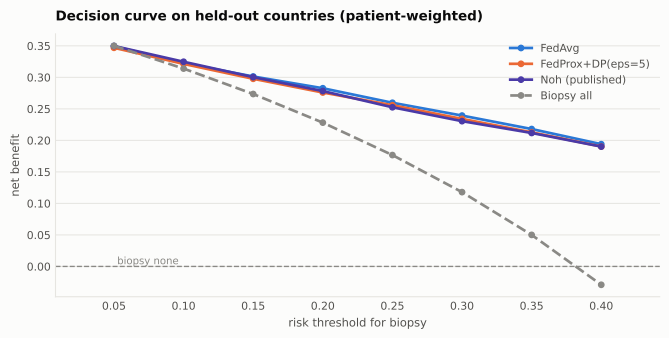


Fig. S3. Decision curves on held-out countries (patient-weighted over the eight leave-one-country-out folds).

significantly from joining and 4 lose. The free-rider learning curve rises from 0.769 with one contributing hospital to 0.815 with five and then flattens (0.830 with 18), so a small consortium suffices, as the base paper argued.

One negative result must be reported. The published Noh calculator, trained in Korea without any federation, is not beaten on the hospital-averaged AUC (FL$-$Noh $= -0.010$ $[-0.016, -0.005]$). The gap is driven by the two smallest Spanish hospitals (16 and 79 patients). On the patient-weighted AUC, FL is ahead (0.856 vs 0.850). FL clearly outperforms the Ettala architecture (FL$-$Ettala $= +0.029$ $[+0.023, +0.035]$).

C. External Validation and Calibration

The base paper evaluates only within its hospital set. We add *leave-one-country-out* validation: models are trained on seven countries and tested on all patients of the eighth (Table S2); preprocessing statistics never include the held-out country. Treating the eight countries as independent external tests, federated training matches centralised training (FedAvg$-$CEN $= -0.000$ $[-0.001, +0.001]$) and the published Noh calculator (FedAvg$-$Noh $= +0.001$ $[-0.011, +0.012]$). DP at $\varepsilon = 5$ costs a small but significant -0.005 $[-0.010, -0.000]$ AUC.

Discrimination transports across countries; calibration does not. Observed/expected ratios of the FedAvg model range from 0.47 to 1.30 across countries, reflecting different csPCa prevalence (12–47%). DP adds a second, previously unreported effect: it makes predictions more extreme. The mean calibration slope is 1.12 $[0.95, 1.30]$ without DP but 0.81 $[0.70, 0.92]$ at $\varepsilon = 5$ and 0.80 $[0.69, 0.91]$ at $\varepsilon = 1$; the last two intervals exclude 1. A simple fix, and one that is already common clinical practice (risk calculators are routinely recalibrated to the local population before use, and some publications omit the intercept for this reason), is *local recalibration* on 20% of a new country's patients (Table S6, Fig. S2). Re-estimating only the intercept reduces $|\log \mathrm{O/E}|$ from 0.263 to 0.071 (paired difference -0.193 $[-0.347, -0.038]$). Re-estimating intercept and slope restores the DP model's slope from 0.81 to 0.99. Neither step changes the AUC, and both use only a few dozen to a few hundred local patients. Decision-curve analysis (Fig. S3) shows positive net benefit over biopsying all men for every federated model at thresholds of 10–40%.

D. Privacy and Drift on Real Hospitals

Table S5 and Fig. S4 report the privacy–utility grid on the real hospitals. With 5,610 patients, privacy is almost free. Even at $\varepsilon = 0.5$ ($\sigma = 97$) the mean per-hospital AUC changes by only -0.005 $[-0.011, +0.001]$. This small cost should not be generalised: it holds for a seven-parameter, regularised logistic model trained on about 4,500 patients. The effect of a given ε depends strongly on the model and the amount of data. For the non-convex federated MLP of the main paper the DP releases are far noisier than for logistic regression (Section VI-G); for simple logistic models even very different ε can give nearly the same result. FedProx reduces client drift by 30.8%, and DP noise inflates drift about tenfold at $\varepsilon = 0.5$. At every privacy level, FedProx equals FedAvg with half the learning rate (largest mean difference 0.001 AUC). FedProx's benefit here is therefore a step-size effect.

E. Stronger Baselines

Table S3 adds SCAFFOLD [6], FedAdam [7], a federated one-hidden-layer MLP, a converged FedAvg (400 instead of 100 local steps) and centralised non-linear references (histogram gradient boosting in the style of LightGBM [8], random forest, MLP). FedNova [9] normalises by the number of local steps and therefore equals FedAvg here, where every hospital runs the same number of steps. Hyper-parameters of the new baselines were tuned on four splits disjoint from the 20 evaluation splits. All federated variants lie within 0.006 AUC of FedAvg within hospitals: FedAdam $+0.002$ $[+0.000, +0.005]$,

TABLE S2
LEAVE-ONE-COUNTRY-OUT EXTERNAL VALIDATION ON PCA-19: TRAIN ON THE OTHER SEVEN COUNTRIES, TEST ON ALL PATIENTS OF THE HELD-OUT COUNTRY (MEAN OF 20 SEEDS). O/E = OBSERVED/EXPECTED EVENTS; SLOPE = CALIBRATION SLOPE (IDEAL 1).

			FedAvg			FedProx + DP ($\varepsilon = 5$)		Noh (published)	
Country	n	csPCa%	AUC	O/E	slope	AUC	slope	AUC	O/E
China	842	31	0.872	0.78	1.08	0.851	0.81	0.874	0.97
Germany	925	43	0.842	1.01	1.07	0.844	0.79	0.833	1.12
Italy	218	12	0.821	0.47	0.89	0.819	0.67	0.825	0.54
Netherlands	266	26	0.904	0.83	1.30	0.877	1.00	0.934	1.03
South Korea	996	33	0.896	0.83	1.45	0.890	1.01	0.886	1.00
Spain	1640	46	0.825	1.24	0.93	0.820	0.72	0.819	1.31
USA	590	47	0.783	1.30	0.86	0.776	0.65	0.765	1.55
United Kingdom	133	29	0.905	0.78	1.10	0.903	0.83	0.907	0.88

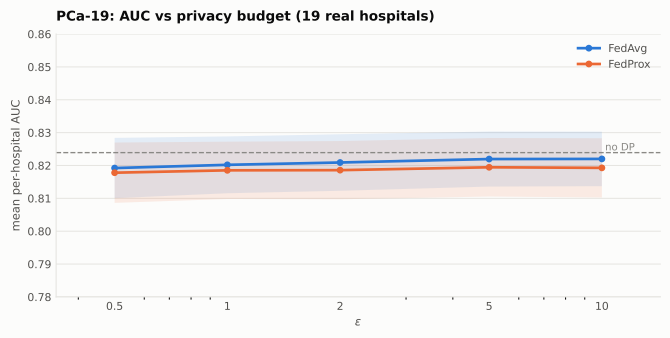


Fig. S4. PCa-19: mean per-hospital AUC vs privacy budget (20 seeds, 95% CI); dashed line = no DP.

federated MLP $+0.005$ $[+0.001, +0.009]$, converged FedAvg $+0.002$ $[+0.001, +0.003]$. SCAFFOLD removes 65% of client drift (FedProx: 31%) without changing AUC (-0.000 $[-0.001, +0.000]$). Under DP at $\varepsilon = 1$ no variant improves on FedAvg (SCAFFOLD $+0.000$ $[-0.000, +0.000]$, FedAdam -0.004 $[-0.008, +0.000]$). Pooled non-linear models gain a little within the training hospitals (gradient boosting $+0.009$ $[+0.004, +0.014]$) but not on unseen countries (gradient boosting -0.003 $[-0.011, +0.004]$, federated MLP $+0.005$ $[-0.000, +0.010]$), where they are also less well calibrated. With four pre-biopsy variables, the information in the data, not the model class or the federated optimiser, sets the ceiling; logistic regression is adequate and the best calibrated.

F. Sensitivity to Training Length

Our default of $T = 20$ rounds and $E = 5$ local epochs (100 gradient steps) was fixed in an earlier phase of the project and keeps the DP noise per step moderate. On PCa-19 it is slightly under-converged for some splits. Repeating the strategy comparison with 400 steps leaves every conclusion unchanged: FL$-$LOC $= -0.005$ $[-0.016, +0.006]$, FR$-$FL $= -0.001$ $[-0.001, -0.001]$, CEN$-$FL $= +0.001$ $[-0.000, +0.003]$, and the published Noh calculator remains ahead on the hospital-averaged AUC (FL$-$Noh $= -0.008$ $[-0.013, -0.003]$) but behind on the patient-weighted AUC (0.858 vs 0.850).

G. Personalisation

Proximal fine-tuning helps exactly where expected. For the 4 hospitals with fewer than 100 patients it improves on the global FedProx model by $+0.039$ $[+0.013, +0.065]$ AUC, and over all hospitals by $+0.010$ $[+0.005, +0.016]$. For large hospitals ($\geq$ 400 patients) it brings no gain ($+0.001$ $[+0.000, +0.001]$), and these hospitals are slightly better served by their own local model (FedProx$-$Local $= -0.005$ $[-0.007, -0.004]$).

H. Subsampled Privacy Accounting

Our main experiments use full-batch updates ($q = 1$), for which no privacy amplification can be claimed. Table S8 replaces them by Poisson-subsampled DP-SGD with sampled-Gaussian RDP accounting. Amplification lowers the noise multiplier needed for $\varepsilon = 1$ from 49.0 (full batch) to 25.6 ($q = 0.3$) and 15.6 ($q = 0.1$). It does not, however, improve utility on held-out splits. On PCa-19 the AUC difference to full batch at $\varepsilon = 1$ is $+0.001$ $[-0.003, +0.004]$ for $q = 0.3$. Smaller expected batches and more steps offset the lower noise multiplier (learning rates for $q < 1$ were tuned on separate splits; a modest gain seen there did not replicate). The conservative full-batch accounting is therefore not what limits utility in this setting. Calibration slopes under DP are equally below 1 with and without subsampling.

I. Retraining-Based Valuation under DP: Full Results

We first follow the standard approach and retrain every coalition with DP at a per-run ε (its real privacy cost is examined in Section VI-C). We computed exact Shapley values over the ten largest hospitals (1,023 coalitions) and, for all 19 hospitals, stratified-sampling Shapley values. Sampling is stratified by coalition size, and the two largest strata (a hospital alone, and leave-one-out) are computed exactly. This needs 1,777 coalitions per seed instead of 2^{19}, and a validation check shows it agrees with exact Shapley to within 0.003 AUC, where plain permutation sampling with 400 permutations errs by 0.014. Rankings are fairly robust under weak privacy but not under strong privacy (Table S7). With ten hospitals, the Spearman correlation with the no-DP ranking is 0.94 at $\varepsilon = 10$, 0.91 at $\varepsilon = 2$, 0.85 at $\varepsilon = 1$ and 0.64 at $\varepsilon = 0.5$, while the AUC changes by less than 0.01 over the same

TABLE S3
STRONGER BASELINES ON PCA-19 (20 HELD-OUT SEEDS). WITHIN-HOSPITAL: MEAN PER-HOSPITAL TEST AUC OVER 19 HOSPITALS. LEAVE-ONE-COUNTRY-OUT: MEAN AUC OVER THE 8 HELD-OUT COUNTRIES AND MEAN CALIBRATION SLOPE; DIFFERENCES VS FEDAVG ARE PAIRED OVER COUNTRIES ($n = 8$). LAST COLUMN: FEDERATED VARIANTS TRAINED WITH DP AT $\varepsilon = 1$. FEDNOVA EQUALS FEDAVG HERE (IDENTICAL LOCAL STEPS AT EVERY HOSPITAL).

Model	Within hospital AUC	−FedAvg [95% CI]	Leave-one-country-out AUC	−FedAvg [95% CI]	Slope	DP $\varepsilon = 1$
FedAvg	0.816	–	0.856	–	1.08	0.811
FedAvg (400 steps)	0.818	$+0.002$ $[+0.001, +0.003]$	0.860	$+0.004$ $[-0.001, +0.009]$	1.02	–
FedProx	0.811	-0.005 $[-0.008, -0.002]$	0.853	-0.003 $[-0.007, +0.000]$	1.12	0.809
SCAFFOLD	0.816	-0.000 $[-0.001, +0.000]$	0.856	$+0.000$ $[-0.001, +0.001]$	1.08	0.811
FedAdam	0.818	$+0.002$ $[+0.000, +0.005]$	0.861	$+0.005$ $[-0.003, +0.012]$	1.07	0.807
Federated MLP	0.821	$+0.005$ $[+0.001, +0.009]$	0.861	$+0.005$ $[-0.000, +0.010]$	0.92	–
Central LR	0.819	$+0.003$ $[+0.000, +0.005]$	0.860	$+0.005$ $[-0.002, +0.011]$	1.02	–
Central gradient boosting	0.824	$+0.009$ $[+0.004, +0.014]$	0.853	-0.003 $[-0.011, +0.004]$	0.91	–
Central random forest	0.820	$+0.004$ $[-0.002, +0.011]$	0.850	-0.006 $[-0.013, +0.001]$	0.95	–
Central MLP	0.828	$+0.012$ $[+0.007, +0.017]$	0.861	$+0.005$ $[-0.001, +0.012]$	1.00	–

range. Because the hospitals' contributions are close together (Shapley values 0.029–0.042 AUC), the identity of the top contributor is fragile even under weak privacy: it keeps its place in 0.75 of runs at $\varepsilon = 10$ and 0.40 at $\varepsilon = 0.5$. With all 19 hospitals the correlation is 0.90 at $\varepsilon = 5$ and 0.76 at $\varepsilon = 1$, and the top contributor keeps its place in 0.45 and 0.45 of runs. Leave-one-out values are near zero for every hospital, so leave-one-out cannot separate contributors (Spearman with Shapley 0.66 [0.56, 0.77]). The instability of contribution rankings under DP therefore appears on real hospitals at strong privacy levels where accuracy is still unaffected, and for the top-ranked position at every privacy level.

J. Membership Inference

The confidence-threshold attack is at chance for every model (AUC 0.500 without DP), while it reaches 0.937 on an overfitted control model. A seven-parameter model trained on about 4,500 patients does not memorise individuals, so the empirical audit again cannot show a DP benefit; DP remains a guarantee against all attacks, including stronger ones not tested here.

S3. DETAILED PCA-19 TABLES

This section collects the detailed tables of the PCa-19 evaluation referred to in the main text.

TABLE S4
PCA-19: THE BASE PAPER'S 19-HOSPITAL PRE-BIOPSY COHORT (5,610 MEN). CSPCA = ISUP GRADE GROUP ≥ 2 AT BIOPSY; MEDIANS OF AGE (YEARS), PSA (NG/ML) AND PROSTATE VOLUME (ML); PI-RADS ≥ 4 IN %.

Hosp.	Country	n	csPCa%	Age	PSA	Vol.	PI-RADS≥4
de1	Germany	763	43	65	8.0	47	74
es7	Spain	711	50	67	7.6	53	55
kr2	South Korea	590	38	68	7.0	42	72
ch2	China	530	33	67	10.3	41	42
kr1	South Korea	406	25	64	6.8	36	28
es6	Spain	346	43	69	7.6	52	78
ch1	China	312	27	66	9.7	45	43
us2	USA	310	48	65	7.3	46	82
us1	USA	280	45	62	5.5	45	62
nl	Netherlands	266	26	67	11.1	56	36
it	Italy	218	12	65	8.0	54	38
es4	Spain	200	34	69	7.6	55	56
es5	Spain	200	47	70	10.6	56	80
de2	Germany	162	47	65	6.5	40	70
gb	United Kingdom	133	29	64	7.6	45	48
es1	Spain	79	42	69	6.3	61	97
es8	Spain	58	38	70	6.5	56	81
es3	Spain	30	60	69	9.0	50	97
es2	Spain	16	38	64	277.0	81	69

S4. PRE-REGISTERED TEST OF THE DECISION RULE

The design and hypotheses below were written before the sweep was run and fixed by their SHA-256 hash (`results/preregistration_nu_sweep.md`, hash `3d5dfcdf...68a1`, 17 Oct 2026, 08:19 UTC); a copy was saved on a second machine at the same time. The registration is internal (a hash, not a public registry such as OSF), and the protocol was written and run on the same day. Its timing therefore cannot be verified by third parties; future tests of the revised rule will be registered publicly (for example on OSF) before they are run. Hypotheses at $\varepsilon = 2$: (H1) OSA−MR > 0 for $\nu \geq 100$; (H2) OSA−MR not positive for $\nu \leq 50$; (H3) OSA−MR increases with ν; (H4) the sign change lies between $\nu = 50$ and 100; (H5) both arms agree at matched ν; (H6) at $\varepsilon = 1$ the switch-over lies at higher ν, or OSA−MR is lower in at least six of eight conditions. Outcome: H1, H3, H5 and H6 supported (H6 through the higher switch-over; OSA−MR was lower at $\varepsilon = 1$ in 5 of 8 conditions); H2 and H4 not supported. No condition, seed or setting was changed after the results were seen. Full results are in Table S14.

S5. CLAIMS AND COMPUTED VERDICTS

Table S15 lists every claim of the study with the evidence and the verdict computed from the result files.

REFERENCES

[1] I. Mironov, K. Talwar, and L. Zhang, "Rényi differential privacy of the sampled Gaussian mechanism," *arXiv preprint arXiv:1908.10530*, 2019.
[2] Y. Zhu and Y.-X. Wang, "Poisson subsampled Rényi differential privacy," in *Proceedings of the 36th International Conference on Machine Learning (ICML)*, ser. Proceedings of Machine Learning Research, vol. 97, 2019, pp. 7634–7642.

TABLE S5
PRIVACY AND DRIFT ON THE 19 REAL HOSPITALS (MEAN PER-HOSPITAL TEST AUC, 20 SEEDS). THE LAST TWO AUC COLUMNS ARE PAIRED DIFFERENCES WITH 95% CI; DRIFT = MEAN $\|w_k - w^{t-1}\|$.

ε	σ	FedAvg	FedProx	FedAvg ($\eta/2$)	FedAvg − no DP	FedProx − FedAvg($\eta/2$)	Drift Avg / Prox
no DP	–	0.824	0.821	0.821	−−	$+0.000$ $[-0.001, +0.002]$	0.33 / 0.23
10	5.7	0.822	0.819	0.819	-0.002 $[-0.007, +0.003]$	$+0.000$ $[-0.001, +0.002]$	0.35 / 0.24
5	10.5	0.822	0.819	0.819	-0.002 $[-0.007, +0.003]$	$+0.000$ $[-0.001, +0.002]$	0.48 / 0.33
2	25.0	0.821	0.819	0.820	-0.003 $[-0.007, +0.001]$	-0.001 $[-0.003, +0.001]$	0.91 / 0.61
1	49.0	0.820	0.819	0.819	-0.004 $[-0.008, +0.001]$	-0.001 $[-0.003, +0.002]$	1.69 / 1.13
0.5	97.0	0.819	0.818	0.819	-0.005 $[-0.011, +0.001]$	-0.001 $[-0.003, +0.001]$	3.30 / 2.19

TABLE S6
LOCAL RECALIBRATION ON 20% OF EACH HELD-OUT COUNTRY, EVALUATED ON THE OTHER 80% (MEAN OVER 8 COUNTRIES AND 20 SEEDS). $|\log \text{O/E}| = 0$ AND SLOPE = 1 ARE IDEAL; DP ROWS ARE FEDPROX + DP.

Model	Recalibration	$\lvert\log$ O/E$\rvert$	Slope	Brier
FedAvg	none	0.263	1.09	0.145
	intercept	0.071	1.09	0.136
	int. + slope	0.074	0.99	0.138
DP $\varepsilon = 5$	none	0.260	0.81	0.150
	intercept	0.080	0.81	0.140
	int. + slope	0.074	0.99	0.140
DP $\varepsilon = 1$	none	0.263	0.80	0.151
	intercept	0.081	0.80	0.140
	int. + slope	0.074	1.00	0.140
Noh	none	0.205	0.97	0.144
	intercept	0.075	0.97	0.137
	int. + slope	0.077	0.97	0.138

TABLE S7
STABILITY OF SHAPLEY CONTRIBUTION RANKINGS UNDER DP ON THE REAL PCA-19 HOSPITALS (20 SEEDS). ρ = SPEARMAN CORRELATION WITH THE NO-DP RANKING; RANK REV. = SHARE OF HOSPITAL PAIRS WHOSE ORDER FLIPS RELATIVE TO THE NO-DP RANKING; TS = PROBABILITY THAT THE NO-DP TOP CONTRIBUTOR STAYS FIRST.

Hospitals	ε	ρ [95% CI]	Rank rev.	TS
10 (exact)	10	0.94 $[0.93, 0.95]$	0.44	0.75
10 (exact)	5	0.94 $[0.92, 0.95]$	0.45	0.75
10 (exact)	2	0.91 $[0.90, 0.93]$	0.57	0.80
10 (exact)	1	0.85 $[0.80, 0.90]$	0.64	0.70
10 (exact)	0.5	0.64 $[0.56, 0.71]$	0.79	0.40
19 (stratified)	5	0.90 $[0.87, 0.93]$	0.80	0.45
19 (stratified)	1	0.76 $[0.71, 0.80]$	0.84	0.45

TABLE S8
POISSON SUBSAMPLING WITH SAMPLED-GAUSSIAN RDP ACCOUNTING (FEDPROX, 20 HELD-OUT SEEDS). σ = NOISE MULTIPLIER NEEDED FOR THE STATED ε OVER THE GIVEN NUMBER OF LOCAL STEPS; PCA-19: MEAN PER-HOSPITAL AUC (DIFFERENCE TO FULL BATCH IN BRACKETS) AND CALIBRATION SLOPE.

Sampling	ε	σ	Steps	AUC	Slope
full batch (q=1)	no DP	–	100	0.811	1.08
full batch (q=1)	10	5.7	100	0.811	0.78
full batch (q=1)	5	10.5	100	0.810	0.78
full batch (q=1)	2	25.0	100	0.810	0.77
full batch (q=1)	1	49.0	100	0.809	0.76
full batch (q=1)	0.5	97.0	100	0.806	0.72
q=0.3	no DP	–	300	0.810 (-0.001)	1.09
q=0.3	10	3.1	300	0.810 (-0.001)	0.79
q=0.3	5	5.6	300	0.810 ($+0.000$)	0.79
q=0.3	2	13.1	300	0.809 (-0.001)	0.79
q=0.3	1	25.6	300	0.810 ($+0.001$)	0.78
q=0.3	0.5	50.5	300	0.809 ($+0.003$)	0.75
q=0.1	no DP	–	1000	0.810 (-0.001)	1.08
q=0.1	10	2.0	1000	0.810 (-0.001)	0.79
q=0.1	5	3.5	1000	0.809 (-0.000)	0.79
q=0.1	2	8.0	1000	0.810 (-0.000)	0.78
q=0.1	1	15.6	1000	0.809 ($+0.000$)	0.77
q=0.1	0.5	30.8	1000	0.807 ($+0.000$)	0.73

TABLE S9
OSA-SHAPLEY ROBUSTNESS ON PCA-19 (TEN HOSPITALS, 20 HELD-OUT SEEDS). TOP: SPEARMAN ρ WITH THE EXACT NON-PRIVATE RANKING. BOTTOM: SHARE OF SPLITS IN WHICH A HOSPITAL WITH RANDOM TRAINING LABELS (ES6) IS RANKED LAST (NO-DP RETRAINING SHAPLEY: 100%).

Setting	$\varepsilon = 2$	$\varepsilon = 1$
Local steps = 10	0.84	0.81
Local steps = 30 (chosen)	0.89	0.80
Local steps = 60	0.90	0.72
Reference set 25% of hospital test data	0.77	0.67
Reference set 50% of hospital test data	0.87	0.77
Useless hospital ranked last		
OSA (cost ε)	100%	95%
MR (cost ε)	70%	80%
OR (cost ε)	100%	100%
Retraining, budget-matched (cost ε)	60%	45%
Retraining, per run (true ε 85 and 33)	100%	100%

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TABLE S10

CONTRIBUTION VALUATION UNDER DP ON THE TEN LARGEST PCA-19 HOSPITALS (20 HELD-OUT SEEDS). AGREEMENT WITH THE EXACT NON-PRIVATE SHAPLEY RANKING (SPEARMAN ρ) AND PROBABILITY THAT THE TRUE TOP CONTRIBUTOR IS RANKED FIRST (TS). *Per-run*: EVERY COALITION RETRAINED AT THE STATED ε, WHOSE TRUE COST FOR EACH HOSPITAL IS THE COMPOSED ε IN COLUMN 2. *Budget-matched*: RETRAINING WITH NOISE CALIBRATED SO THAT THE TOTAL COST IS THE STATED ε. OSA-SHAPLEY COSTS THE STATED ε ONCE.

	True ε of	Spearman ρ with exact Shapley			TS		
ε	per-run retraining	per-run	budget-matched	OSA	per-run	budget-matched	OSA
10	988	0.94	0.47	**0.90**	0.75	0.45	**0.60**
5	334	0.94	0.32	**0.90**	0.70	0.35	**0.60**
2	85	0.90	0.18	**0.89**	0.65	0.10	**0.55**
1	33	0.76	0.10	**0.80**	0.55	0.10	**0.40**
0.5	14	0.46	0.02	**0.54**	0.50	0.05	**0.35**

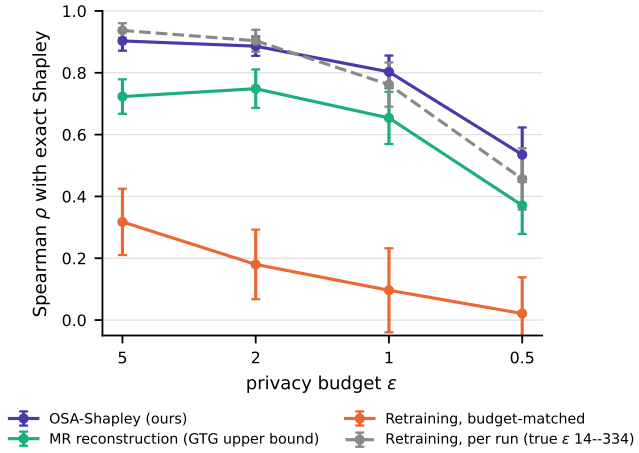


Fig. S5. Agreement of each valuation with the exact non-private Shapley ranking of the ten largest PCa-19 hospitals (20 held-out splits, 95% CI). OSA-Shapley, MR reconstruction and budget-matched retraining all cost the stated ε; per-run retraining costs the composed ε of Table S10.

TABLE S11

VALUATION WITH A NON-CONVEX MODEL: A ONE-HIDDEN-LAYER MLP (16 TANH UNITS, WEIGHT DECAY 10^{-3}) TRAINED BY FEDAVG. SPEARMAN ρ WITH THE EXACT NON-PRIVATE RETRAINING SHAPLEY RANKING OF THE TEN LARGEST PCA-19 HOSPITALS (20 HELD-OUT SPLITS; BUDGET-MATCHED RETRAINING ON 5 OF THEM). OSA-ENS.: COALITION PREDICTION = SIZE-WEIGHTED AVERAGE OF THE MEMBERS' LOG-ODDS; OSA-PAR.: SIZE-WEIGHTED PARAMETER AVERAGE FROM A SHARED INITIALISATION; MR: EXACT MULTI-ROUND RECONSTRUCTION FROM ONE DP FEDAVG-MLP RUN. ALL COLUMNS COST THE STATED ε.

ε	OSA-ens.	OSA-par.	MR	OSA-ens.$-$MR	Budget-m.
none	0.91	0.67	0.78	$+0.13$ $[+0.07, +0.19]$	–
5	0.68	0.52	0.82	-0.14 $[-0.19, -0.09]$	–
2	0.65	0.65	0.82	-0.17 $[-0.25, -0.09]$	0.16
1	0.60	0.60	0.78	-0.18 $[-0.29, -0.07]$	0.05
0.5	0.42	0.44	0.74	-0.32 $[-0.46, -0.18]$	–

TABLE S12

VALUATIONS THAT COST THE PRIVACY BUDGET ONCE (SPEARMAN ρ WITH THE EXACT NON-PRIVATE RETRAINING SHAPLEY RANKING). OSA: ONE DP MODEL PER HOSPITAL. MR: EXACT MULTI-ROUND RECONSTRUCTION FROM ONE DP-FEDAVG RUN (GTG-SHAPLEY IS ITS TRUNCATED MONTE-CARLO APPROXIMATION). OR: ONE-ROUND RECONSTRUCTION FROM THE SAME RUN. ALL THREE ARE POST-PROCESSING OF RELEASES THAT ALREADY COST ε. OSA$-$MR: PAIRED DIFFERENCE WITH 95% CI. BOTH GROUPS ARE REAL PCA-19 HOSPITALS, 20 HELD-OUT SPLITS; NO HOSPITAL IS SIMULATED OR PARTITIONED.

	Ten largest hospitals (266–763 patients)				Ten smallest hospitals (16–266 patients)			
ε	OSA	MR	OR	OSA$-$MR	OSA	MR	OR	OSA$-$MR
none	**0.96**	0.74	0.42	$+0.22$ $[+0.18, +0.27]$	**0.95**	0.86	0.79	$+0.10$ $[+0.05, +0.14]$
5	**0.90**	0.72	0.53	$+0.18$ $[+0.14, +0.22]$	**0.85**	0.83	0.81	$+0.03$ $[-0.00, +0.06]$
2	**0.89**	0.75	0.51	$+0.14$ $[+0.08, +0.19]$	**0.79**	0.78	0.74	$+0.00$ $[-0.06, +0.06]$
1	**0.80**	0.65	0.50	$+0.15$ $[+0.03, +0.26]$	**0.68**	0.74	0.59	-0.06 $[-0.15, +0.03]$
0.5	**0.54**	0.37	0.39	$+0.16$ $[+0.04, +0.29]$	–	–	–	–

TABLE S13

EVIDENCE BEHIND THE DECISION RULE, FROM TWO REAL FEDERATIONS. p: MODEL PARAMETERS; ν: MEDIAN TRAINING PATIENTS PER HOSPITAL DIVIDED BY $\sqrt{p}$, WHICH SETS THE SIGNAL-TO-NOISE RATIO OF ONE DP RELEASE (PROPOSITION 3). ρ: SPEARMAN CORRELATION WITH THE EXACT NON-PRIVATE RETRAINING SHAPLEY RANKING (20 HELD-OUT SPLITS). OSA$-$MR: PAIRED DIFFERENCE WITH 95% CI AT $\varepsilon = 2$. SIZE: AGREEMENT OF HOSPITAL SIZE WITH THE SAME RANKING.

Federation	Model	K	p	ν	OSA no DP	OSA $\varepsilon = 2$	MR $\varepsilon = 2$	OSA$-$MR	Size	Best at $\varepsilon = 2$
PCa-19, 10 largest	logistic	10	7	114	0.96	0.89	0.75	$+0.14$ $[+0.08, +0.19]$	0.08	OSA
PCa-19, 10 smallest	logistic	10	7	45	0.95	0.79	0.78	$+0.00$ $[-0.06, +0.06]$	0.74	OSA, MR or size
Heart Disease, 4	logistic	4	21	43	0.92	0.15	0.45	-0.30 $[-0.59, -0.01]$	-0.38	MR
PCa-19, 10 largest	MLP	10	129	26	0.91	0.65	0.82	-0.17 $[-0.25, -0.09]$	–	MR

TABLE S14

PRE-REGISTERED TEST OF THE DECISION RULE ON THE TEN LARGEST PCA-19 HOSPITALS (20 HELD-OUT SPLITS). ARM A KEEPS A RANDOM FRACTION OF EVERY HOSPITAL'S TRAINING PATIENTS; ARM B ADDS UNINFORMATIVE FEATURES. IN EVERY CONDITION THE REFERENCE IS THE EXACT NON-PRIVATE RETRAINING SHAPLEY RANKING OF THAT CONDITION. ρ: SPEARMAN CORRELATION WITH THE REFERENCE; OSA$-$MR: PAIRED DIFFERENCE WITH 95% CI.

			$\varepsilon = 2$			$\varepsilon = 1$		
Condition	p	ν	OSA	MR	OSA$-$MR	OSA	MR	OSA$-$MR
A: 100% of patients	7	113	0.89	0.75	$+0.14$ $[+0.08, +0.19]$	0.80	0.65	$+0.15$ $[+0.03, +0.26]$
A: 60% of patients	7	68	0.80	0.72	$+0.08$ $[+0.00, +0.16]$	0.65	0.51	$+0.14$ $[+0.02, +0.26]$
A: 40% of patients	7	45	0.74	0.55	$+0.18$ $[+0.08, +0.29]$	0.48	0.36	$+0.12$ $[-0.04, +0.27]$
A: 25% of patients	7	28	0.58	0.46	$+0.12$ $[+0.00, +0.24]$	0.40	0.36	$+0.04$ $[-0.07, +0.16]$
A: 15% of patients	7	17	0.52	0.61	-0.09 $[-0.20, +0.03]$	0.35	0.53	-0.18 $[-0.32, -0.03]$
B: +21 features	28	57	0.78	0.71	$+0.07$ $[-0.03, +0.18]$	0.66	0.54	$+0.13$ $[+0.01, +0.24]$
B: +50 features	57	40	0.75	0.70	$+0.05$ $[-0.02, +0.12]$	0.64	0.62	$+0.01$ $[-0.08, +0.10]$
B: +100 features	107	29	0.75	0.71	$+0.05$ $[-0.02, +0.11]$	0.60	0.68	-0.07 $[-0.15, +0.00]$

TABLE S15

CLAIM–EVIDENCE SUMMARY. VERDICTS ARE COMPUTED FROM THE RESULT FILES: 'SUPPORTED' ONLY IF THE 95% CI OF THE PAIRED DIFFERENCE OVER SEEDS EXCLUDES 0 IN THE CLAIMED DIRECTION; THE PRE-REGISTERED ROWS ARE SCORED BY THE CRITERIA FIXED IN ADVANCE (SECTION S4).

Area	Claim	Evidence (mean [95% CI])	Verdict
PCa-19 (19 hospitals)	Joining FL gives hospitals a significant average local gain	FL - LOC (mean per-hospital AUC) = −0.005 [-0.015, +0.006]	not significant
PCa-19 (19 hospitals)	Free-riders get (almost) the same model as participants	FR - FL = −0.002 [-0.002, −0.001]	supported
PCa-19 (19 hospitals)	Federated training matches pooled training	CEN - FL = +0.002 [-0.001, +0.004]	supported
PCa-19 (19 hospitals)	FL beats the published Noh calculator (hospital-averaged AUC)	FL - Noh = −0.010 [-0.016, −0.005]; patient-weighted FL 0.856 vs Noh 0.850	opposite
PCa-19 (19 hospitals)	Our re-implementation reproduces the base paper's code exactly	noh: max |difference| 1.2e-14 over 5x19 cells (50 repeats); ettala: max |difference| 7.8e-15 over 5x19 cells (50 repeats)	supported
PCa-19 (19 hospitals)	Federated model transports to unseen countries as well as pooled training	leave-one-country-out, FedAvg - centralised = −0.000 [-0.001, +0.001] (8 countries)	supported
PCa-19 (19 hospitals)	DP lowers the calibration slope (predictions more extreme) on held-out countries	slope at ε=5 = 0.81 [0.70, 0.92] over 8 countries (no DP 1.12)	supported
PCa-19 (19 hospitals)	Local recalibration (20% of a new country) fixes calibration-in-the-large	|log O/E| recalibrated - transported = −0.193 [-0.347, −0.038]	supported
PCa-19 (19 hospitals)	Strong privacy (ε = 0.5) costs < 0.02 AUC at this scale	FedAvg(ε=0.5) - no DP = −0.005 [-0.011, +0.001]	supported
PCa-19 (19 hospitals)	FedProx's effect equals halving FedAvg's learning rate	largest |FedProx - FedAvg(lr/2)| over 6 privacy levels = 0.0009 AUC; drift reduction 30.8%	explained by step size
PCa-19 (19 hospitals)	Personalisation helps small hospitals (< 100 patients)	PFL - FedProx = +0.039 [+0.013, +0.065]	supported
PCa-19 (19 hospitals)	DP noise destabilises Shapley rankings of real hospitals	10 hospitals: Spearman 0.94 (ε=10) → 0.64 (ε=0.5)	supported
PCa-19 (19 hospitals)	OSA-Shapley beats retraining at the same real privacy cost (ε = 2)	Spearman with exact Shapley: OSA - budget-matched retraining = +0.71 [+0.58, +0.84]	supported
PCa-19 (19 hospitals)	OSA-Shapley is not significantly worse than per-run retraining, which spends far more privacy (ε = 2)	OSA - per-run retraining = −0.02 [-0.06, +0.03] (per-run real ε = 85)	supported
PCa-19 (19 hospitals)	OSA-Shapley beats reconstruction from the DP training run (MR, GTG upper bound) at the same cost (ε = 2)	OSA - MR = +0.14 [+0.08, +0.19]; OSA - OR = +0.37 [+0.26, +0.48]	supported
PCa-19 (19 hospitals)	OSA-Shapley ranks a hospital with random labels last more often than budget-matched retraining (ε = 2)	ranked last: OSA 100%, MR 70%, budget-matched retraining 60%	supported
PCa-19 (19 hospitals)	OSA-Shapley beats sharing hospital size (10 largest hospitals, ε = 2)	OSA - size = +0.80 [+0.70, +0.91]; rho size 0.08, statistics −0.18	supported
PCa-19 (19 hospitals)	OSA-Shapley beats the single-model AUC of the same DP releases (10 largest, ε = 1)	OSA - single-model AUC = +0.10 [+0.05, +0.14] (ε = 1); +0.04 [-0.00, +0.08] (ε = 2)	supported
PCa-19 (19 hospitals)	On the 10 smallest real hospitals OSA-Shapley beats reconstruction (ε = 2)	OSA - MR = +0.00 [-0.06, +0.06]; OSA - size = +0.05 [-0.01, +0.10]	not significant
PCa-19 (19 hospitals)	OSA-Shapley also beats reconstruction with a non-convex model (federated MLP, ε = 2)	OSA-ensemble - MR = −0.17 [-0.25, −0.09] (rho 0.65 vs 0.82)	opposite
PCa-19 (19 hospitals)	With an MLP, post-processing valuation beats budget-matched retraining (ε = 2)	OSA-ensemble - budget-matched = +0.47 [-0.34, +1.28] on 5 splits; MR 0.80	not significant
PCa-19 (19 hospitals)	A stronger federated algorithm or federated MLP beats converged FedAvg	best: Federated MLP +0.005 [+0.001, +0.009] vs FedAvg; converged FedAvg (400 steps) +0.002 [+0.001, +0.003]	not significant
PCa-19 (19 hospitals)	Non-linear models (gradient boosting) transport better to unseen countries	leave-one-country-out AUC, gradient boosting - FedAvg = −0.003 [-0.011, +0.004] (8 countries); slope 0.91	not significant
PCa-19 (19 hospitals)	SCAFFOLD reduces client drift more than FedProx without changing AUC	drift FedAvg 0.322, FedProx 0.223, SCAFFOLD 0.113; AUC SCAFFOLD - FedAvg = −0.000 [-0.001, +0.000]	supported
PCa-19 (19 hospitals)	Subsampled DP (q = 0.3) improves utility over full batch at ε = 1	mean per-hospital AUC difference = +0.001 [-0.003, +0.004]	not significant
Heart (4 hospitals)	Without DP, OSA-Shapley beats reconstruction in a second real federation	OSA - MR = +0.22 [+0.05, +0.39] (rho 0.92 vs 0.70)	supported
Heart (4 hospitals)	Under DP, OSA-Shapley also beats reconstruction in the heart federation (ε = 2)	OSA - MR = −0.30 [-0.59, −0.01] (rho 0.15 vs 0.45); about 43 patients per parameter	opposite
Heart (4 hospitals)	Hospital size identifies contributions (no DP)	rho size −0.38; OSA - size = +1.30 [+1.12, +1.48]	not supported
Pre-registered rule test	Pre-registered H1: OSA beats MR wherever $\nu \geq 100$ (ε = 2)	OSA - MR at ν = 113: +0.14 [+0.08, +0.19]	supported
Pre-registered rule test	Pre-registered H2: OSA stops beating MR at $\nu \leq 50$ (ε = 2)	OSA - MR at ν = 45: +0.18 [+0.08, +0.29]; ν = 28: +0.12 [+0.00, +0.24]	not supported
Pre-registered rule test	Pre-registered H3: OSA - MR increases with ν	Spearman(ν, OSA - MR) = 0.57	supported
Pre-registered rule test	Pre-registered H4: the switch-over lies between ν = 50 and 100	observed switch-over at ν = 21	not supported
Pre-registered rule test	Pre-registered H5: fewer patients and more parameters agree at matched ν	ν = 45 (fewer patients) +0.18 [+0.08, +0.29] vs ν = 40 (more parameters) +0.05 [-0.02, +0.12]	supported
Pre-registered rule test	Pre-registered H6: stronger privacy (ε = 1) moves the switch-over up	ε = 1 lower in 5 of 8 conditions; switch-overs at ν = 26, 29, 38	supported